Mathematics – I (M-I)
GTU # 3110014

Dedicated Faculty, Committed Education
Darshan
Institute of Engineering & Technology

Unit – 5
Multiple integral

Prof. Kartikraj Dhamsaniya
Humanities & Science Department
Darshan Institute of Engineering & Technology, Rajkot

kartikraj.dhamsaniya@darshan.ac.in
972 41 42 400

Sr. No.	Content	Total Hrs	% Weightage
05	Multiple integral, Double integral over Rectangles and general regions, double integrals as volumes, Change of order of integration, double integration in polar coordinates, Area by double integration, Triple integrals in rectangular, cylindrical and spherical coordinates, Jacobian, multiple integral by substitution.	08	20 %

Unit 5 Multiple Integral

- ✓ **Double integral over Rectangles and general regions**
- ✓ Double integrals as volumes
- ✓ Change of order of integration
- ✓ Double integration in polar coordinates
- ✓ Area by double integration
- ✓ **Triple integrals in rectangular**
- ✓ Cylindrical and spherical coordinates
- ✓ Jacobian
- ✓ Multiple integral by substitution

METHOD – 6 : D.I. OVER GENERAL REGION IN POLAR COORDINATES(1)

METHOD – 5 : D.I. BY CHANGE OF ORDER OF INTEGRATION(3)
METHOD – 13 : D.I. BY CHANGE OF VARIABLE IN CARTESIAN COORDINATES
METHOD – 14 : D.I. BY CHANGE OF VARIABLE IN POLAR COORDINATES(2)

METHOD – 7 : AREA BY DOUBLE INTEGRATION IN CARTESIAN COORDINATES(1)
METHOD – 8 : AREA BY DOUBLE INTEGRATION IN POLAR COORDINATES
METHOD – 4 : DOUBLE INTEGRALS AS VOLUMES(1)

METHOD – 9 : T.I. OVER GENERAL REGION IN CARTESIAN COORDINATES
METHOD – 10 : T.I. OVER GENERAL REGION IN CYLINDRICAL COORDINATES
METHOD – 11 : T.I. OVER GENERAL REGION IN SPHERICAL COORDINATES(1)

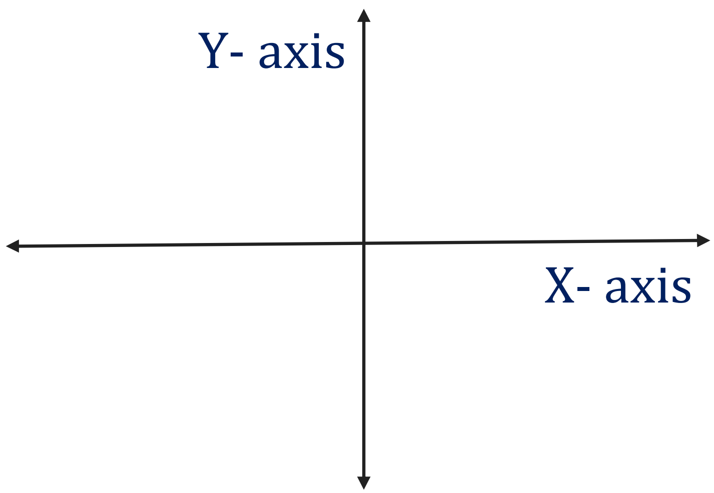
Outline

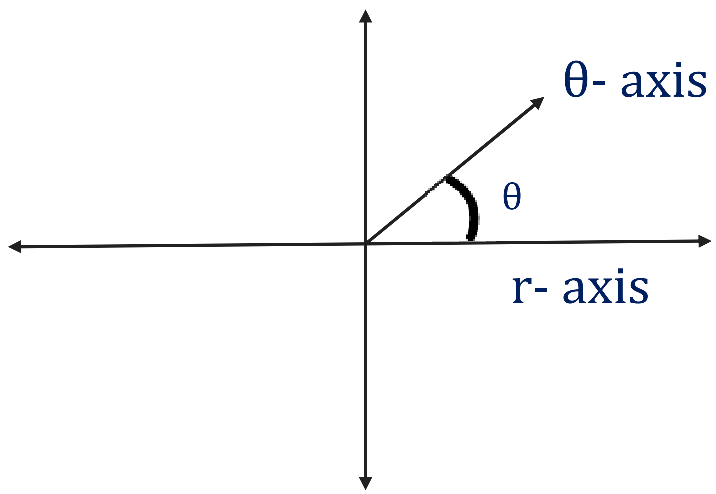
- D.I. over general region – PC
- D.I. by Change of Order of integration
- D.I. by Change of Variable of integration – CC & PC
- Area by D.I. – CC & PC
- Volume by D.I.
- T.I. over General Region

D.I. over general region – PC (Method-6)

Remark

- In Cartesian coordinate, we have x-axis & y-axis (in two dimension).
- While in polar coordinate, we have r-axis & θ-axis.





Strip parallel to X-axis

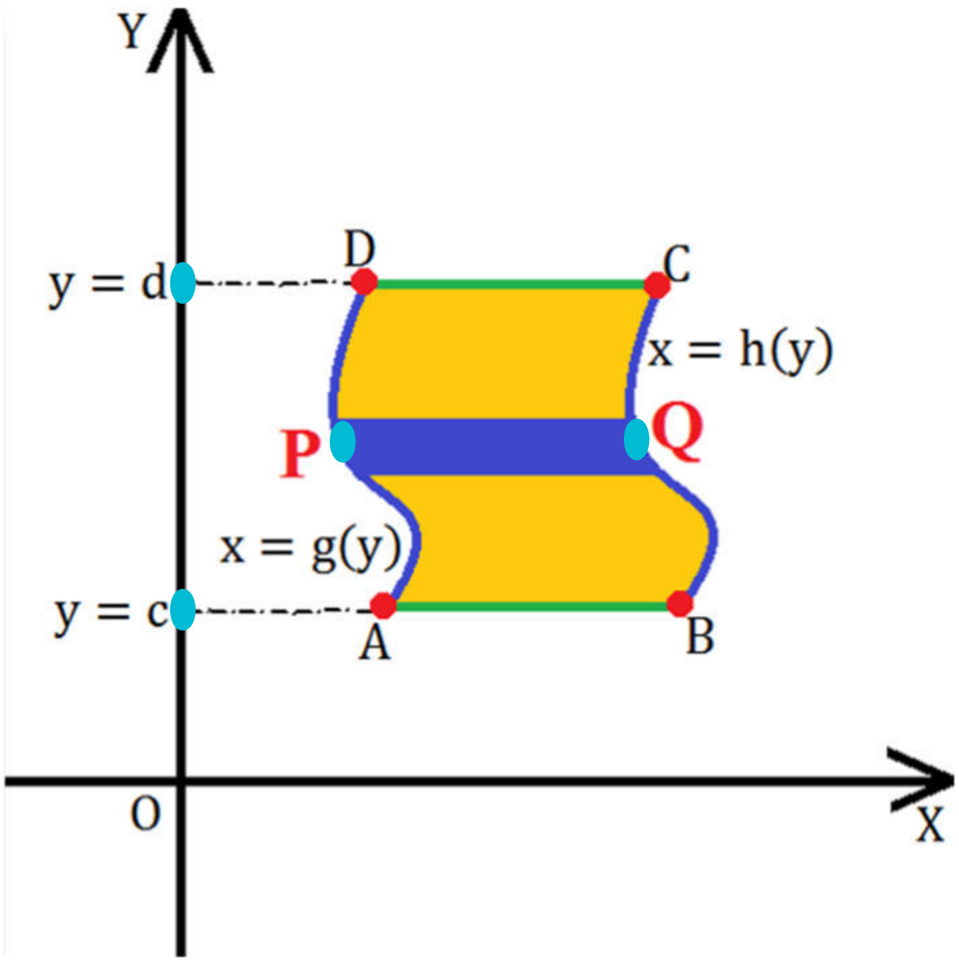
So if we take a strip parallel to **X-axis** then

✓ we have to integrate given function

first with respect to variable **x** and

then variable y.

$$\iint_R f(x, y)\, dA = \int_{y=c}^{y=d} \left\{ \int_{x=g(y)}^{x=h(y)} f(x, y)\, dx \right\} dy$$



Strip parallel to Y-axis

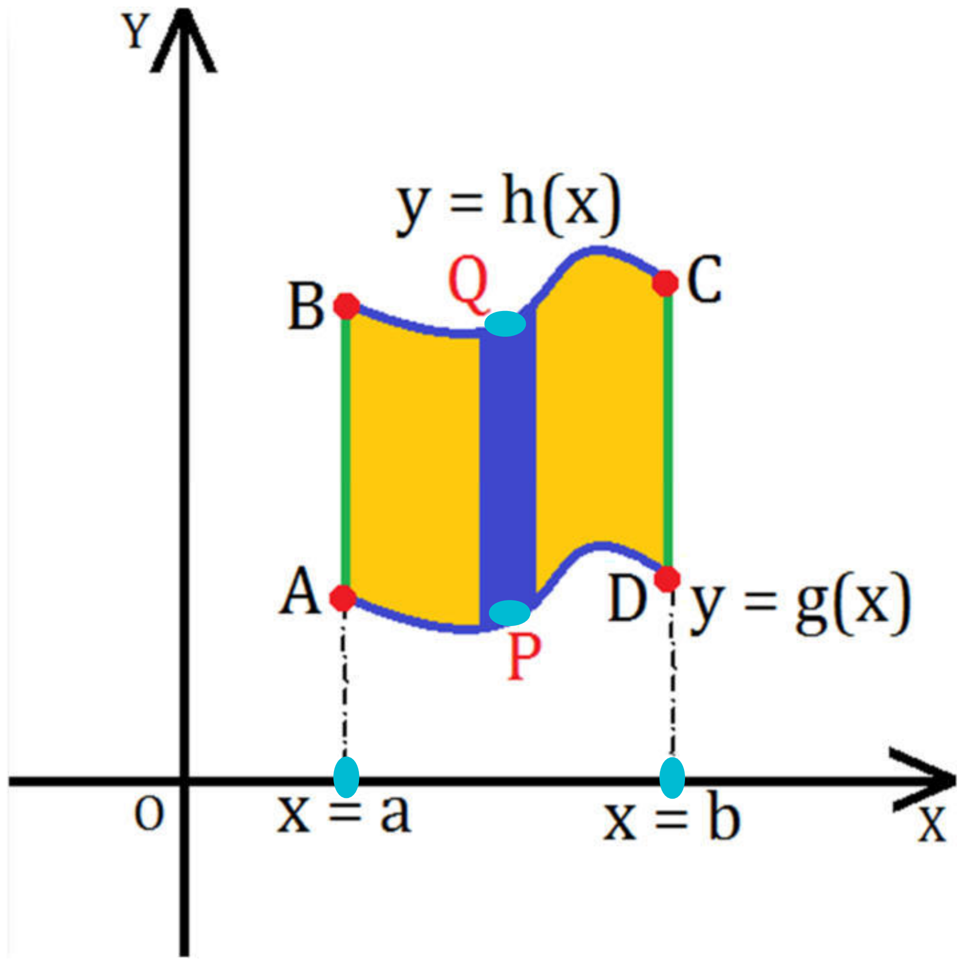
So if we take a strip parallel to **Y-axis** then

- ✓ we have to integrate given function

    first with respect to variable **y** and

    then variable x.

$$\iint_R f(x,y)\, dA = \int_{x=a}^{x=b} \left\{ \int_{y=g(x)}^{y=h(x)} f(x,y)\, dy \right\} dx$$



Procedure to find double integral over General Region - PC

- If we want to find the value of given double integral over general region in PC, we will follow following steps:

1. Draw all the curves and identify required region.
2. Take the strip from origin within the region.
3. If required then divide the region into two parts.
4. Find limits of inner integral from the strip i.e. r_1 & r_2
5. Find limits of outer integral from angle i.e. θ_1 & θ_2
6. Calculate the example as the example of double integrals.

Strip

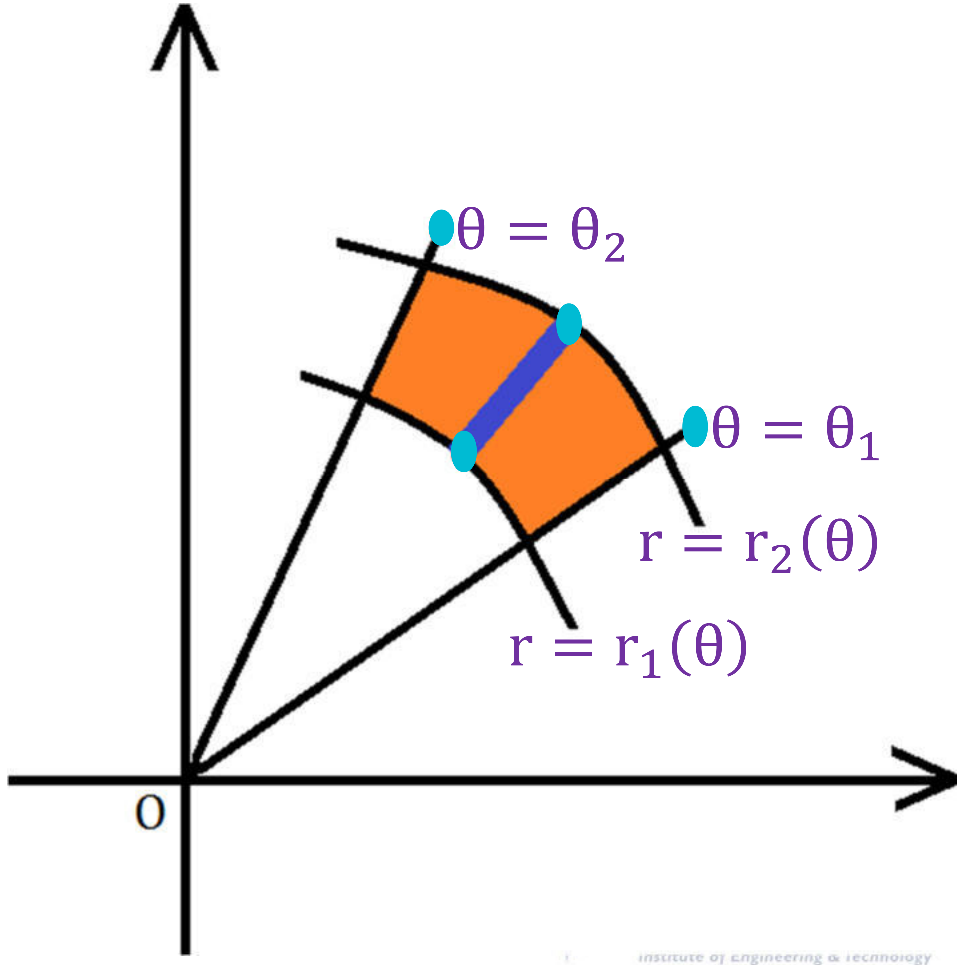
$$\iint_R f(r,\theta)\,dA = \int_{\theta=\theta_1}^{\theta=\theta_2}\left\{\int_{r=r_1(\theta)}^{r=r_2(\theta)} f(r,\theta)\,\boxed{dr}\right\}d\theta$$

The **lower limit** of inner integral is obtained from the curve where the strip starts

i.e. $r = r_1(\theta)$ and

The **upper limit** of inner integral is obtained from the curve where the strip ends

i.e. $r = r_2(\theta)$.



Few Important Curves in PC

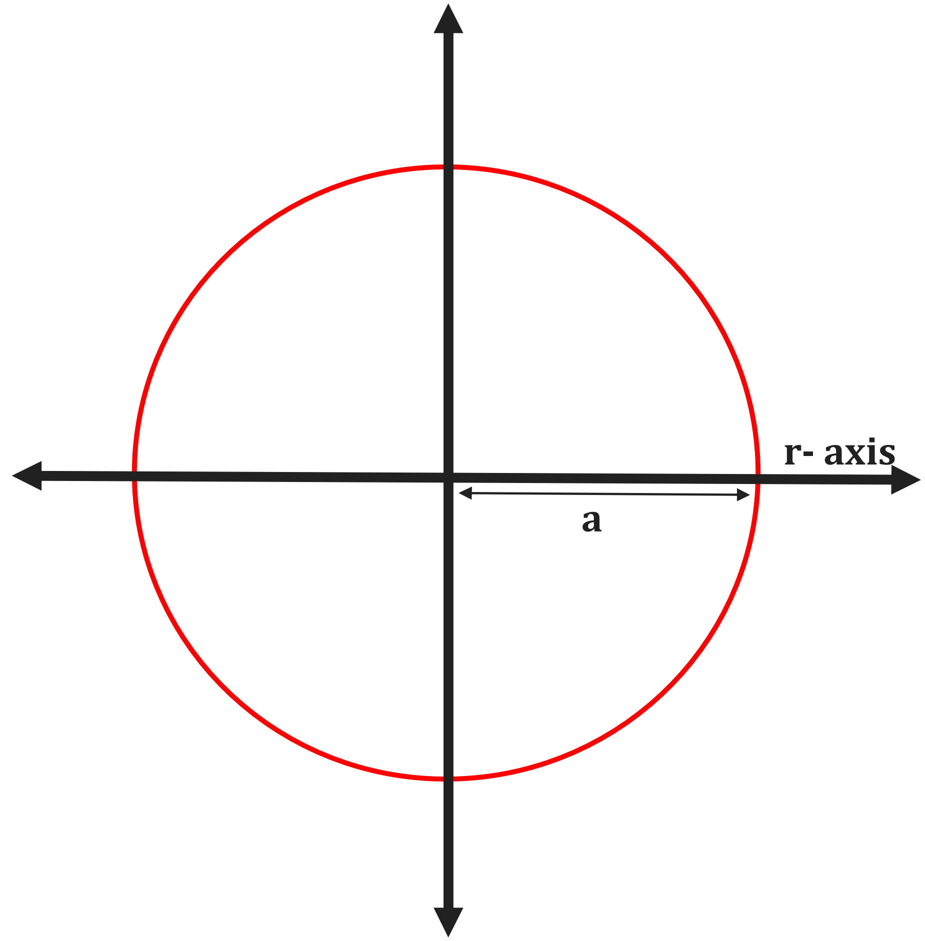
Circle, Cardioid, Lemniscates

Equation of Circle

$r = a$

Radius : a

Centre : (0, 0)

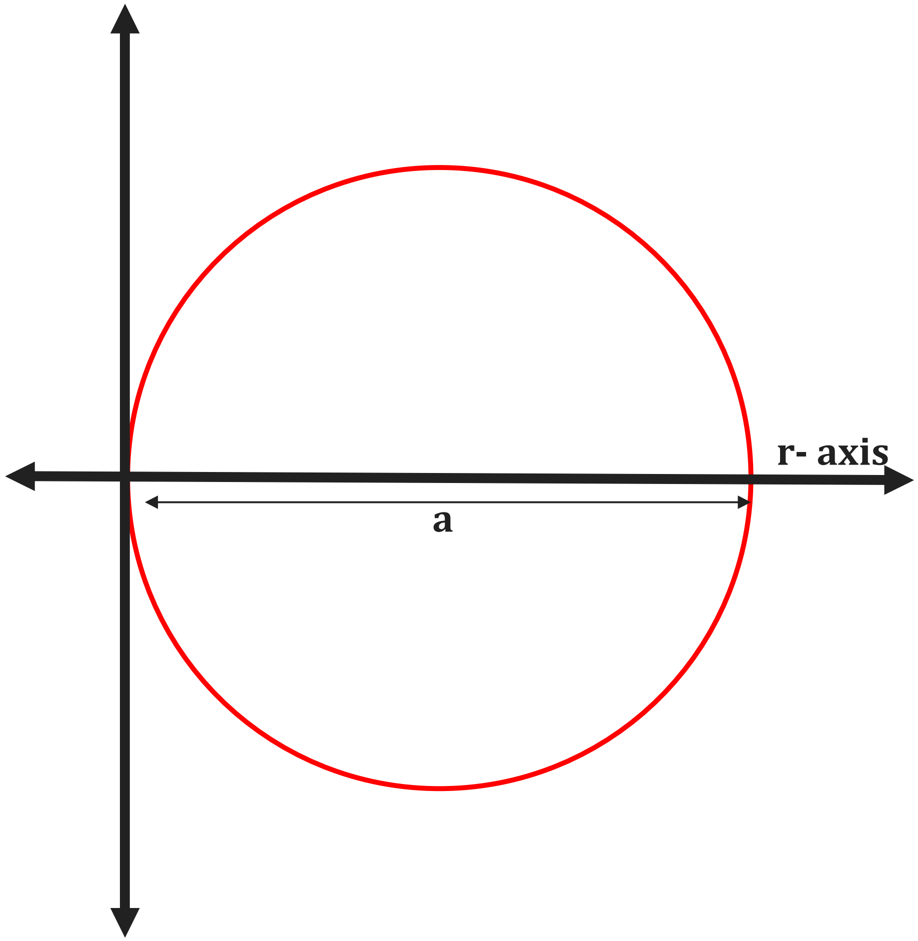


Equation of Circle

$$r = a\cos\theta$$

Radius : $\dfrac{a}{2}$

Centre : $\left(\dfrac{a}{2}, 0\right)$

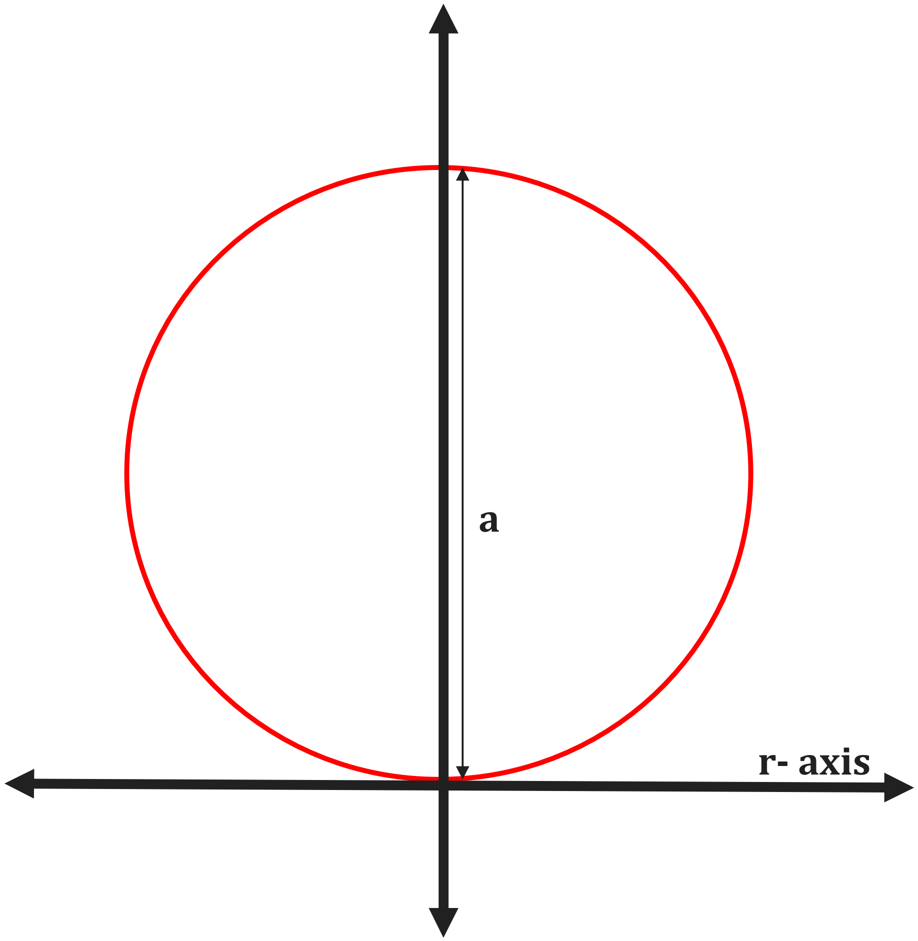


Equation of Circle

$r = a\sin\theta$

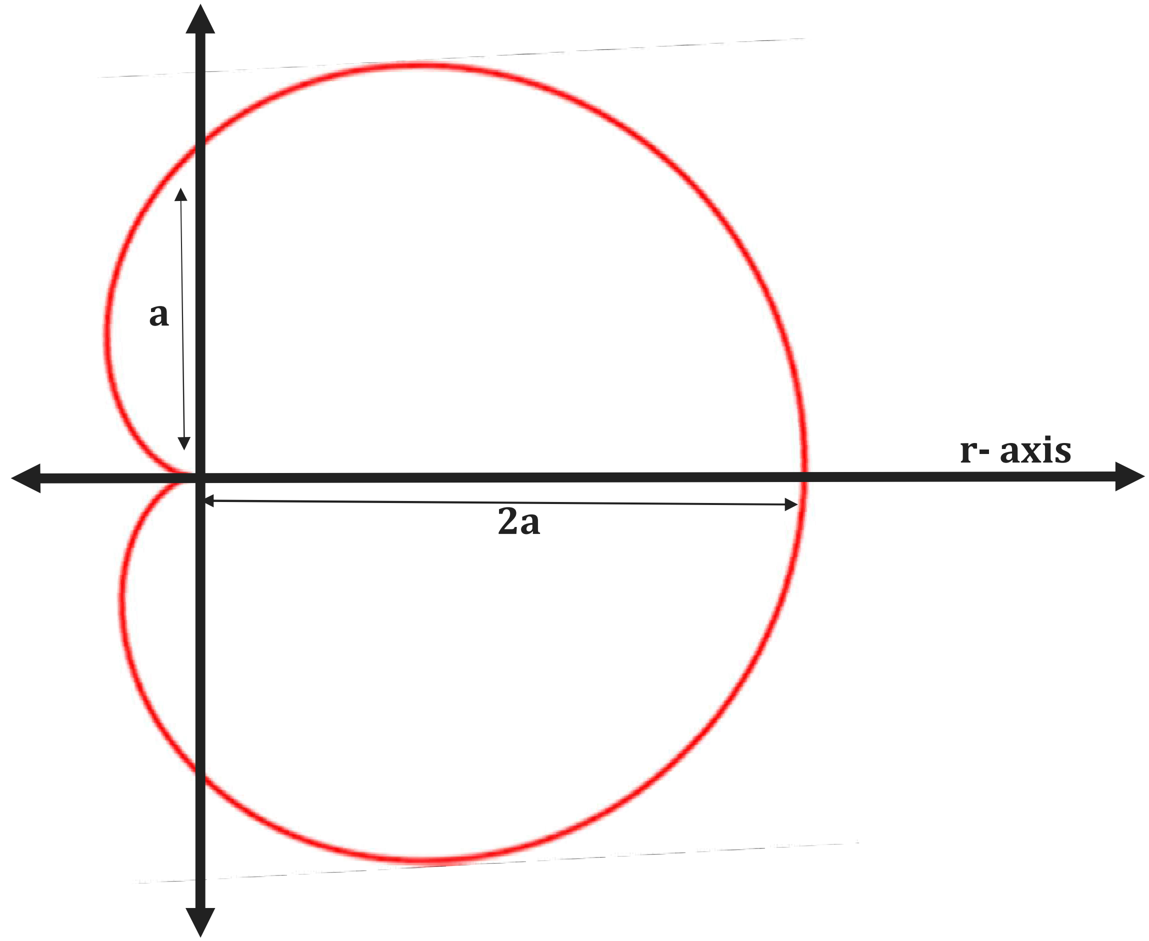
Radius $: \dfrac{a}{2}$

Centre $: \left(0, \dfrac{a}{2}\right)$



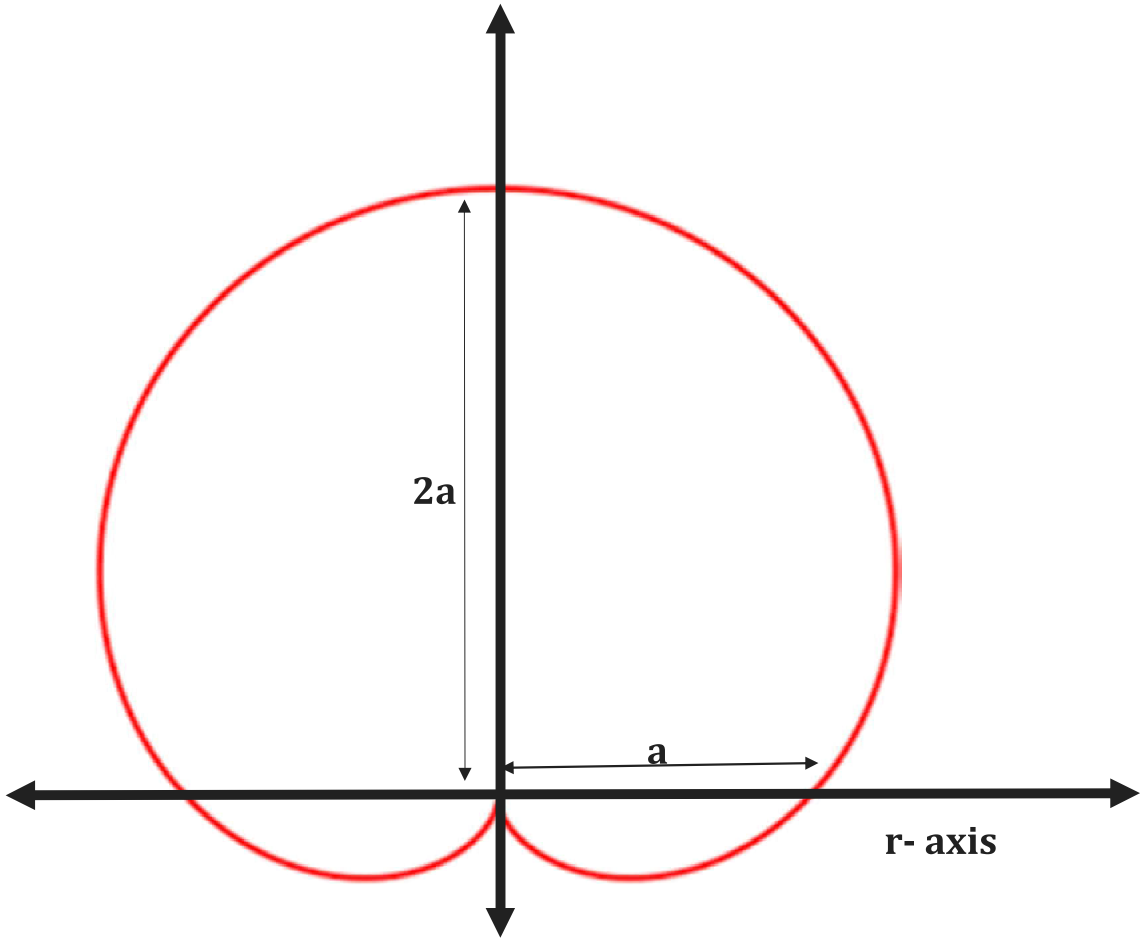
Equation of Cardioid

$r = a(1 + \cos\theta)$



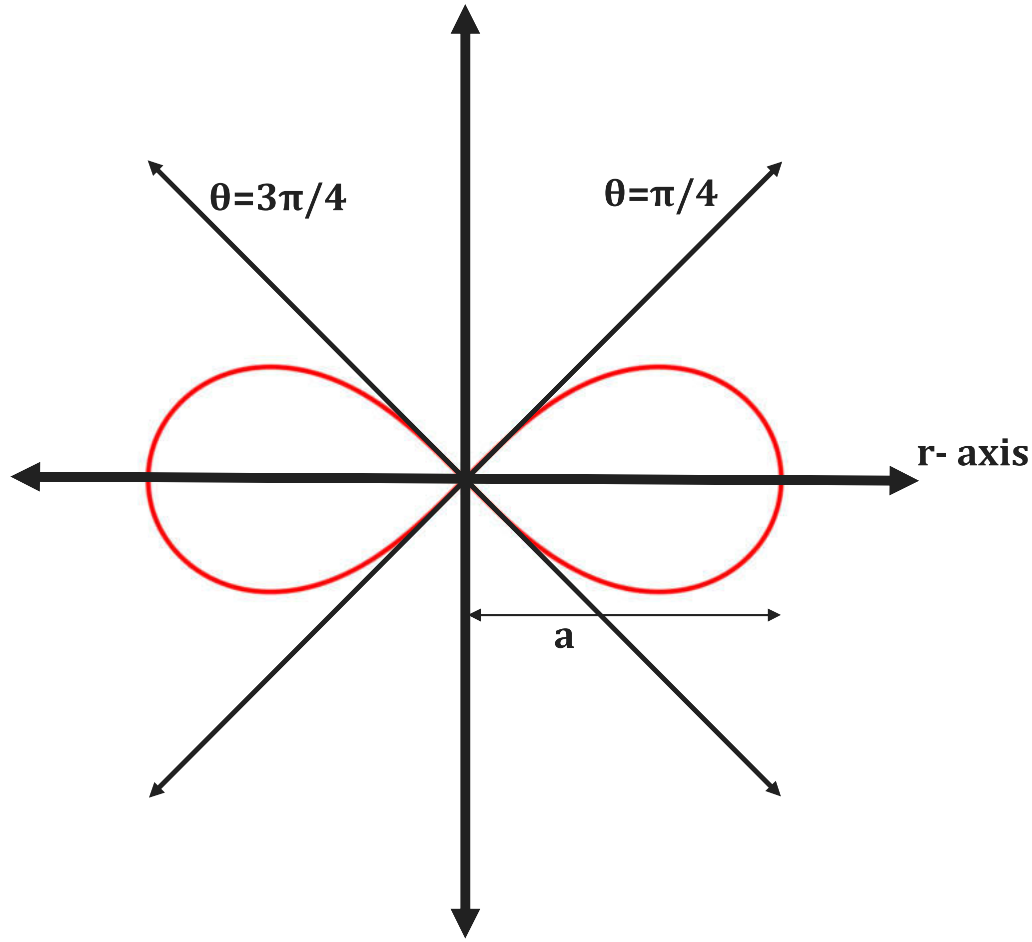
Equation of Cardioid

$r = a(1 + \sin\theta)$



Equation of Lemniscates

$r^2 = a^2 \cos 2\theta$



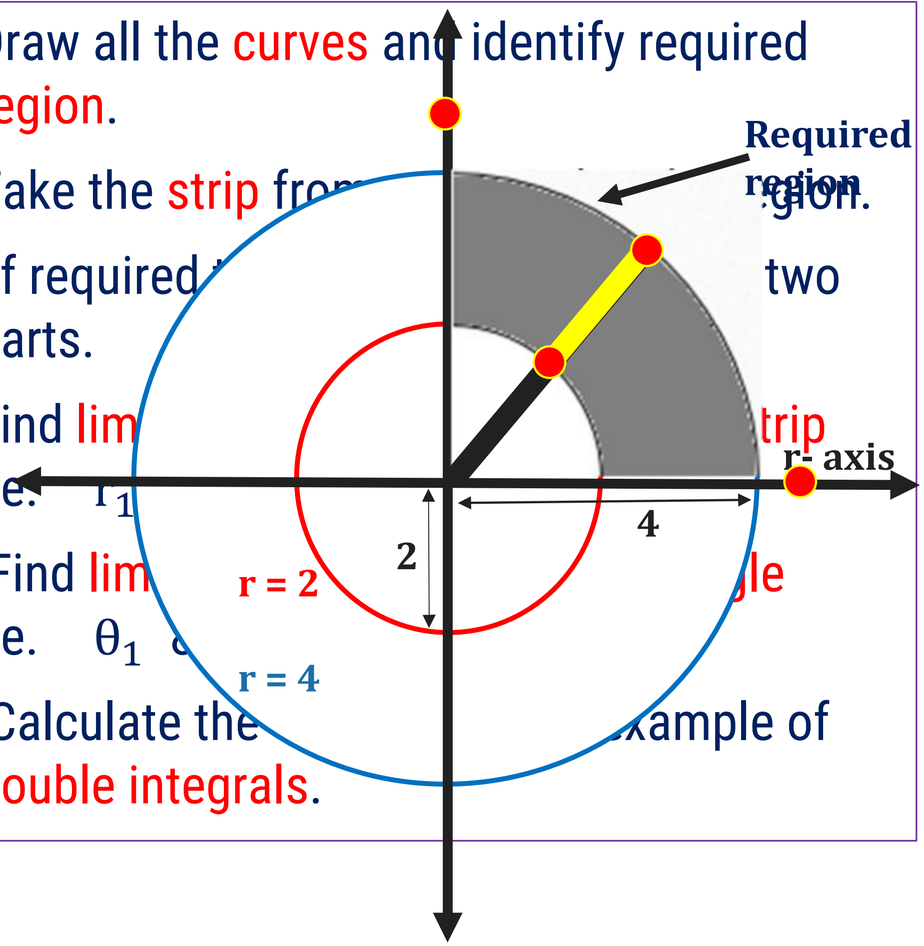
Find the value of $\iint r^3 \sin 2\theta \, dr \, d\theta$ over the area bounded in the first quadrant between the circles $r = 2$ and $r = 4$.

Solution:

$$I = \int_{\theta=0}^{\pi/2} \int_{r=2}^{4} r^3 \sin 2\theta \, \mathbf{dr} \, d\theta$$

……

- Draw all the curves and identify required region.
- Take the strip fro… …gion.
- If required … two parts.
- Find lim… …trip i.e. r_1 …
- Find lim… …gle i.e. θ_1 …
- Calculate the … …xample of double integrals.

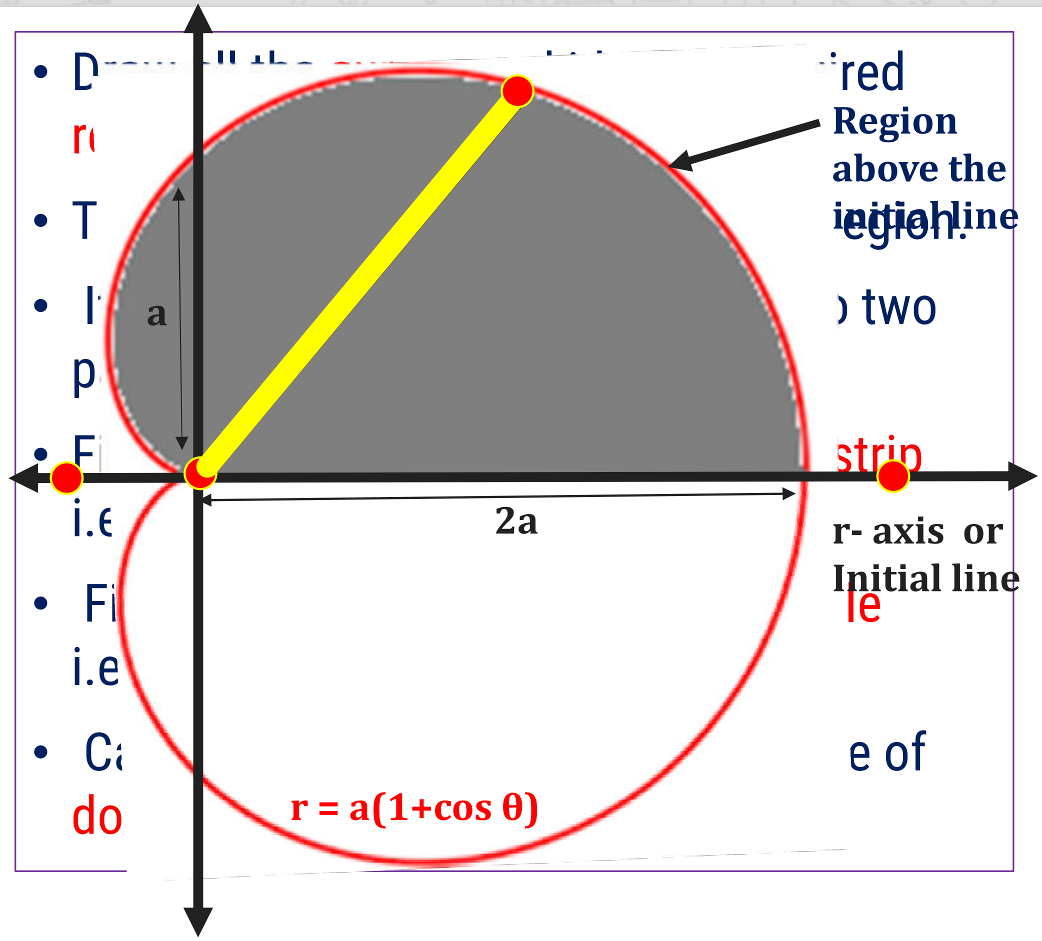


Method 6 ⇒ Example – 4

Evaluate $\iint r\sin\theta \, dr \, d\theta$ over the area of the cardioid $r = a(1+\cos\theta)$ and above the initial line.

Solution:

$$I = \int_{\theta=0}^{\pi} \int_{r=0}^{a(1+\cos\theta)} r \sin\theta \, \mathbf{dr} \, d\theta$$



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D.I. by change of Order of integration(Method-5)

Procedure to find double integral by change of Order

- If we want to find the value of given double integral by changing the order of Integration, we will follow following steps:

1. Draw the region from given data i.e. from limits of integrals.
2. Draw the strip as per given limit of integrals.
3. Now remove the given strip and draw the new strip perpendicular to old strip.
4. Find the limit of both integrals from new strip & region.
5. Calculate the example as the example of double integrals.

Change the order of

$$\int_{1}^{2} \int_{1-\sqrt{2x-x^2}}^{1+\sqrt{2x-x^2}} f(x,y)\, dy\, dx \,.$$

Solution:

$$I = \int_{1}^{2} \int_{1-\sqrt{2x-x^2}}^{1+\sqrt{2x-x^2}} f(x,y)\, dy\, dx$$

- Draw the region from given data i.e. from limits of integrals.
- Draw the strip as per given limit of integrals.
- Now remove the given strip and draw the new strip perpendicular to old strip.
- Find the limit of both integrals from new strip & region.
- Calculate the example as the example of double integrals.

From the limits of Integral, We have

$$y = 1 - \sqrt{2x - x^2}$$

$$y = 1 + \sqrt{2x - x^2}$$

$$x = 1 \quad \& \quad x = 2$$

$$\therefore (x-1)^2 + (y-1)^2 = 1$$

$$x = 1 \quad \& \quad x = 2$$

Rough Work

$$y = 1 - \sqrt{2x - x^2}$$

$$y - 1 = -\sqrt{2x - x^2}$$

$$(y-1)^2 = \left(-\sqrt{2x - x^2}\right)^2$$

$$(y-1)^2 = 2x - x^2$$

$$(y-1)^2 = \underline{2x - x^2 - 1} + 1$$

$$(y-1)^2 = -(x^2 - 2x + 1) + 1$$

$$(y-1)^2 = -(x-1)^2 + 1$$

$$(x-1)^2 + (y-1)^2 = 1$$

$$y = 1 + \sqrt{2x - x^2}$$

$$y - 1 = +\sqrt{2x - x^2}$$

$$(y-1)^2 = \left(+\sqrt{2x - x^2}\right)^2$$

$$(y-1)^2 = 2x - x^2$$

$$(y-1)^2 = 2x - x^2 - 1 + 1$$

$$(y-1)^2 = -(x^2 - 2x + 1) + 1$$

$$(y-1)^2 = -(x-1)^2 + 1$$

$$(x-1)^2 + (y-1)^2 = 1$$

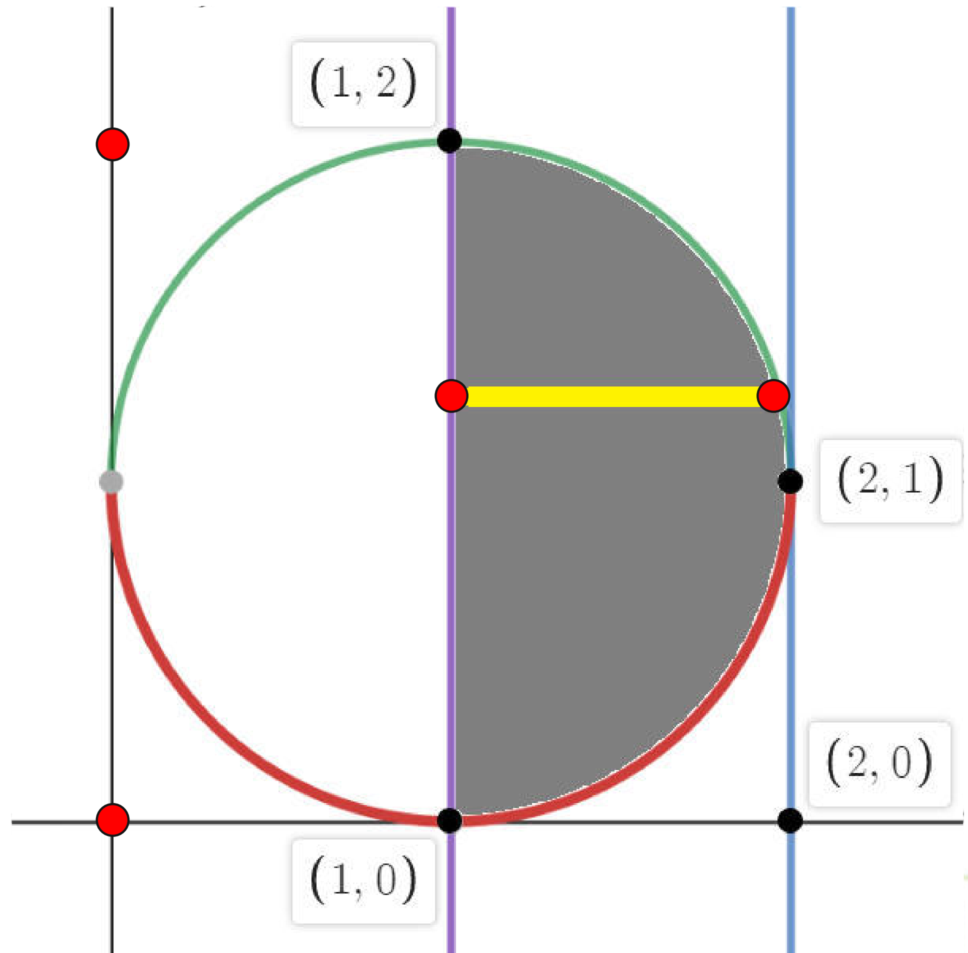
Method 5 $\Rightarrow$ Example – 4(continue)

- We have

$$y = 1 - \sqrt{2x - x^2}$$
$$y = 1 + \sqrt{2x - x^2}$$
$$x = 1 \;\&\; x = 2$$

$$\therefore I = \int_{1}^{2} \int_{1-\sqrt{2x-x^2}}^{1+\sqrt{2x-x^2}} f(x,y)\, dy\, dx$$

$$= \int_{y=0}^{2} \int_{x=1}^{1+\sqrt{2y-y^2}} f(x,y)\, \mathbf{dx}\, dy$$



Evaluate $\int_{0}^{3}\int_{\sqrt{\frac{x}{3}}}^{1} e^{y^3}\,dy\,dx.$

Solution: S-19 (7mark)

$$I = \int_{0}^{3}\int_{\sqrt{\frac{x}{3}}}^{1} e^{y^3}\,dy\,dx$$

- From the limits of Integral, We have

$$y = \sqrt{\frac{x}{3}}$$

$$y = 1$$

$$x = 0 \quad \& \quad x = 3$$

$$\therefore x = 3y^2 \quad \& \quad y = 1$$

$$x = 0 \quad \& \quad x = 3$$

- We have

$$y = \sqrt{x/3} \quad \& \quad y = 1$$

$$x = 0 \quad \& \quad x = 3$$

$$\therefore I = \int_{0}^{3} \int_{\sqrt{\frac{x}{3}}}^{1} e^{y^3}\, dy\, dx$$

$$= \int_{0}^{1} \int_{x=0}^{3y^2} e^{y^3}\, \mathbf{dx}\, dy$$

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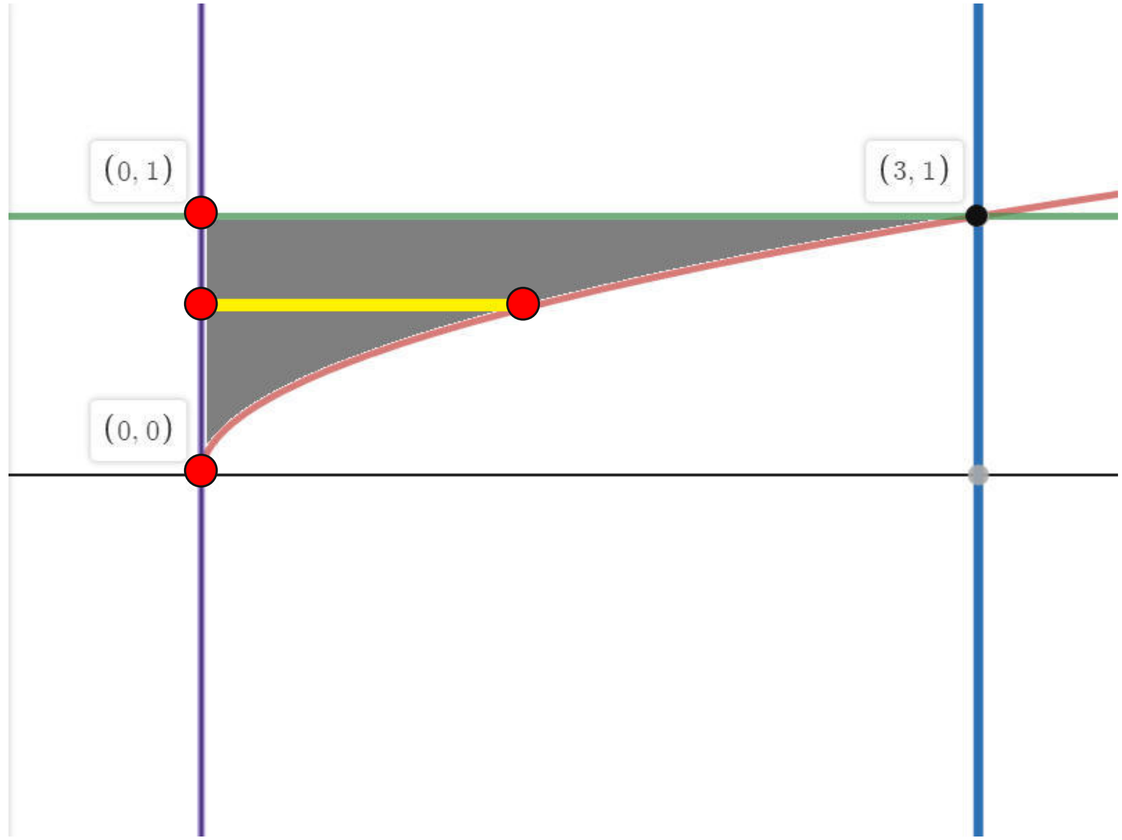


Method 5 ⇒ Example – 12

Change the order of integrations and hence evaluate

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}}\, dy\, dx.$$

Solution:

$$I = \int_0^1 \int_0^{\sqrt{1-x^2}} \frac{e^y}{(e^y+1)\sqrt{1-x^2-y^2}}\, dy\, dx$$

- From the limits of Integral, We have

$$y = \sqrt{1-x^2}$$

$$y = 0$$

$$x = 0 \quad \& \quad x = 1$$

$$\therefore x^2 + y^2 = 1 \quad \& \quad y = 0$$

$$x = 0 \quad \& \quad x = 1$$

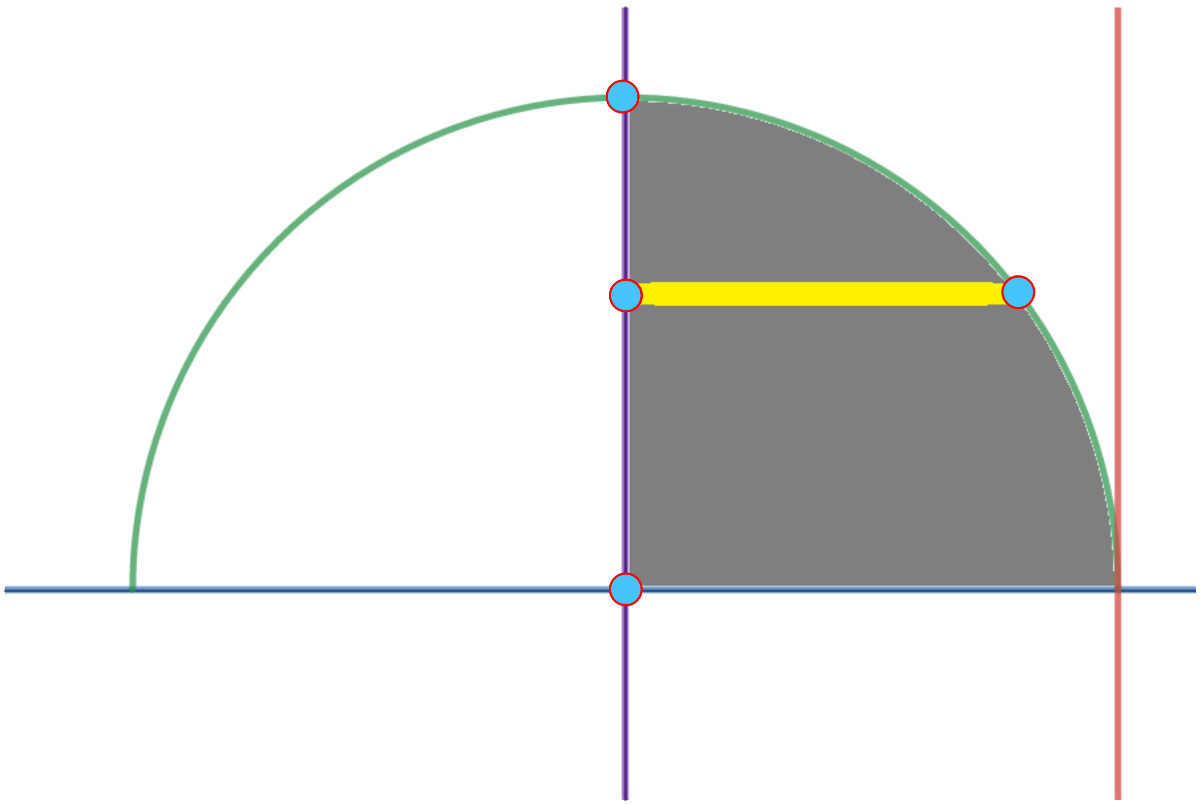
- We have

$$y = \sqrt{1 - x^2} \quad \& \quad y = 0$$

$$x = 0 \quad \& \quad x = 1$$

$$\therefore I = \int_0^1 \int_0^{\sqrt{1-x^2}} ???\, dy\, dx$$

$$= \int_0^1 \int_{x=0}^{\sqrt{1-y^2}} \frac{e^y}{(e^y + 1)\sqrt{1 - x^2 - y^2}}\, dx\, dy$$



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Method 5 ⇒ Example – 17

Change the order of integration and evaluate

$$\int_0^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} dy\, dx.$$

Solution:

$$I = \int_0^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} dy\, dx$$

- From the limits of Integral, We have

$$y = \frac{x^2}{4} \quad \& \quad y = 2\sqrt{x}$$

$$x = 0 \quad \& \quad x = 4$$

$$\therefore x^2 = 4y, \quad y^2 = 4x$$

$$x = 0 \quad \& \quad x = 4$$

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Method 5 ⇒ Example – 17(continue)

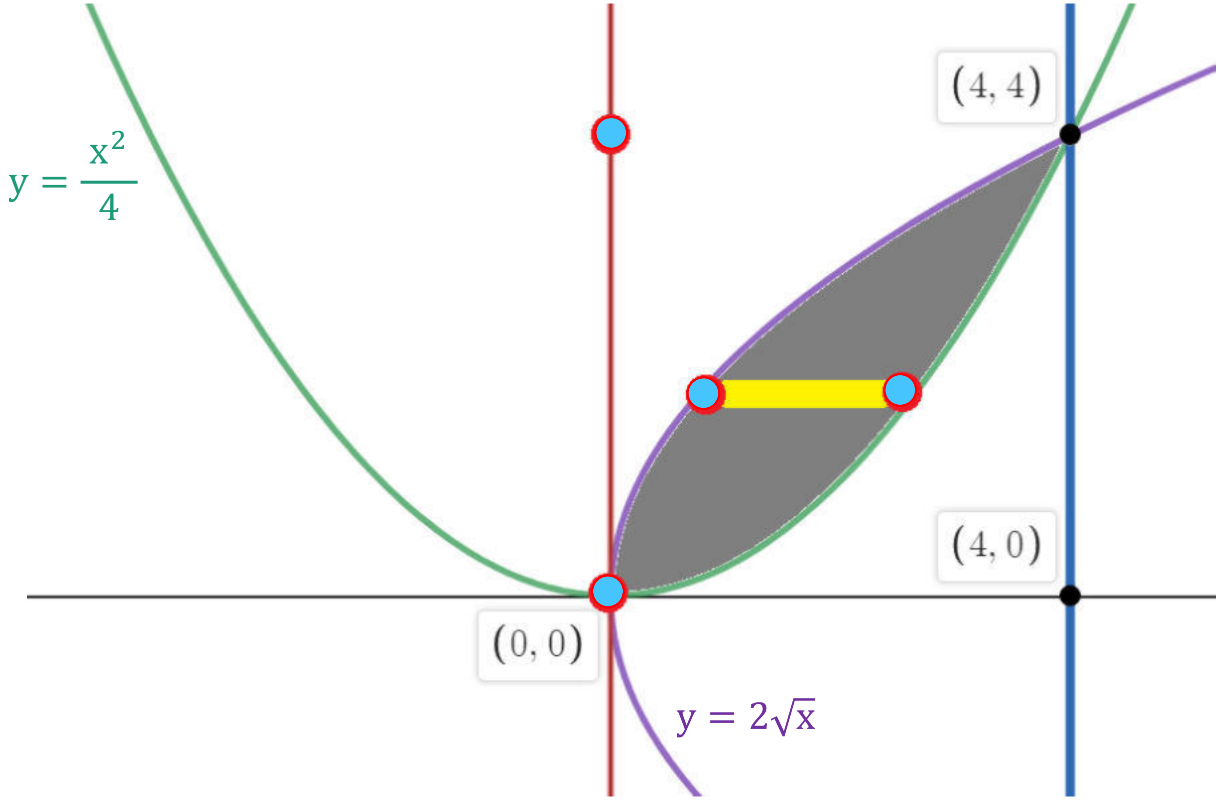
- We have

$$y = x^2/4 \quad \& \quad y = 2\sqrt{x}$$

$$x = 0 \quad \& \quad x = 4$$

$$\therefore I = \int_{0}^{4} \int_{x^2/4}^{2\sqrt{x}} 1 \, dy \, dx$$

$$= \int_{0}^{4} \int_{x = y^2/4}^{2\sqrt{y}} 1 \, \mathbf{dx} \, dy$$



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Method 5 ⇒ Example – 19

Express as single integral and hence evaluate

$$\int_0^1 \int_0^y (x^2 + y^2)\, dx\, dy + \int_1^2 \int_0^{2-y} (x^2 + y^2)\, dx\, dy$$

Solution:

- From the limits of 1st Integral, We have

$$x = 0,\ x = y \qquad \& \qquad y = 0,\ y = 1.$$

- From the limits of 2nd Integral, We have

$$x = 0,\ x = 2 - y \qquad \& \qquad y = 1,\ y = 2.$$

- We have

$x = 0, \quad x = y,$

$y = 0, \quad y = 1$

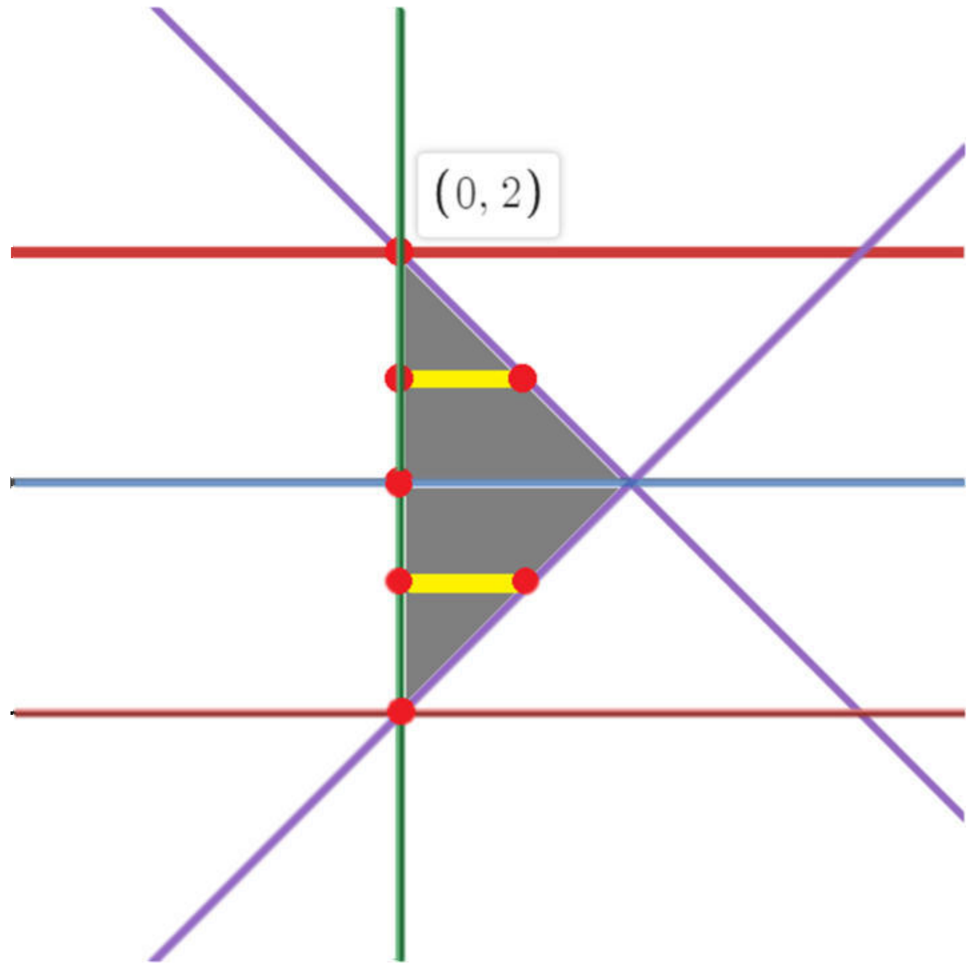
&

$x = 0, \quad x = 2 - y$

$y = 1, \quad y = 2$

$$\int_0^1 \int_0^y (x^2 + y^2)\, dx\, dy$$

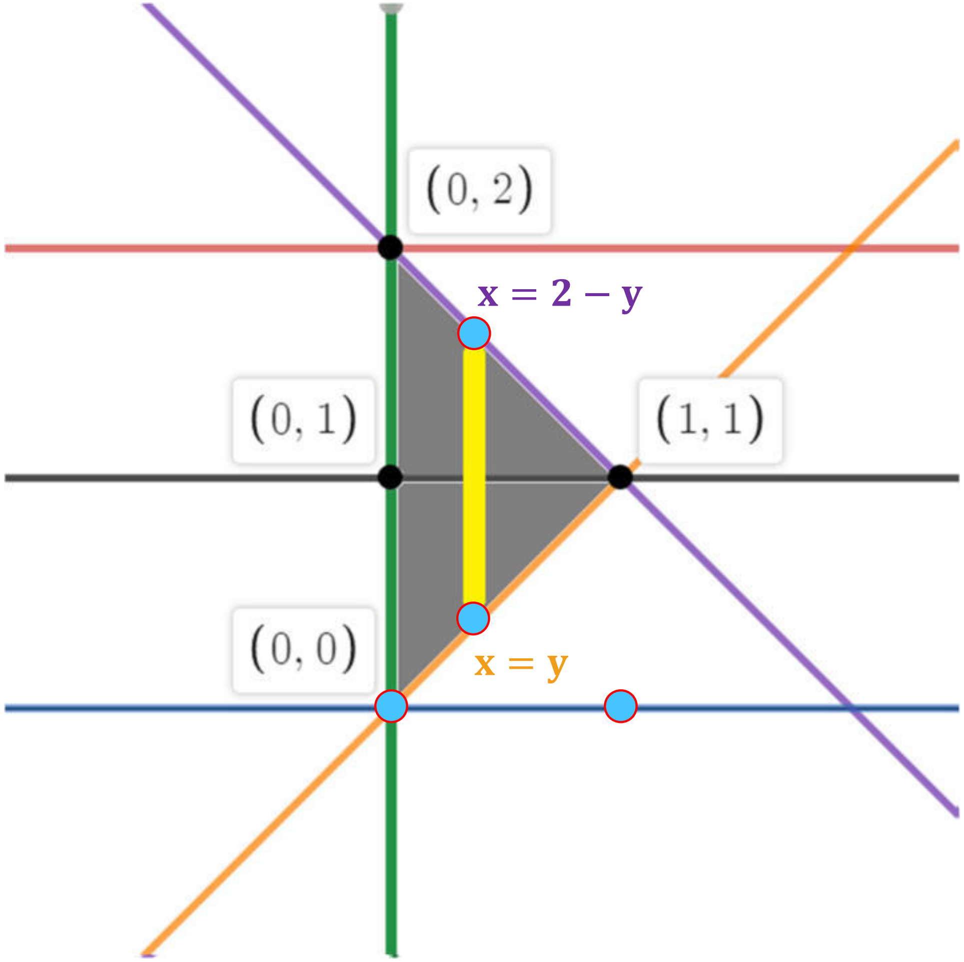
$$\int_1^2 \int_0^{2-y} (x^2 + y^2)\, dx\, dy$$



Method 5 ⇒ Example – 19(continue)

$$I = \int_0^1 \int_0^y (x^2 + y^2)\, dx\, dy + \int_1^2 \int_0^{2-y} (x^2 + y^2)\, dx\, dy$$

$$= \int_0^1 \int_{y=x}^{2-x} (x^2 + y^2)\, \mathbf{dy}\, dx$$



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KEEP
LEARNING
AND
ENJOY